भारतीय मानक

Indian Standard

IS 18698 : 2026

डाइमेथाइल ईथर (डीएमई) मिश्रित तरलीकृत पेट्रोलियम गैस (एलपीजी) — विशिष्टि

(पहला पुनरीक्षण)

Dimethyl Ether (DME) Blended Liquefied Petroleum Gas (LPG) — Specification

(First Revision)

ICS 71.080.60; 78.160.30.

भारतीय मानक ब्यूरो
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May 2026 **Price Group 6**

Petroleum and their Related Products of Synthetic or Biological or Natural Origin Sectional Committee, PCD 03

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Petroleum and their Related Products of Synthetic or Biological or Natural Origin Sectional Committee had been approved by the Petroleum, Coal and Related Products Division Council.

Liquefied Petroleum Gas (LPG) is being used in India to cater to the energy needs of domestic, commercial and industrial sectors apart from use as an automotive fuel. The consumption of LPG in the country is expected to rise. This necessitates the use of alternate fuels to partially substitute LPG with fuel such as synthetically produced dimethyl ether (DME).

DME is an alternative which can be blended with LPG. It is the simplest ether with oxygen connecting two methyl groups having no C-C bond. DME can be blended with LPG and the manufacturer/supplier shall comply with household cooking standards as per the following:

a) IS 4246 : 2025 — Domestic gas stove and built in hob for use with LPG — Specification

b) IS 8737 : 2017 — Valve fittings for use with liquefied petroleum gas (LPG) of upto 250 litre water capacity — Specification (*second revision*)

c) IS 9573 (Part 1) : 2017 — Rubber hose for liquefied petroleum gas (LPG) — Specification: Part 1 Industrial application

d) IS 9573 (Part 2) : 2017 — Rubber hose for liquefied petroleum gas (LPG) — Specification: Part 2 Domestic and commercial application

e) IS 9798 : 2013 — Low pressure regulators for use with liquefied petroleum gas (LPG) — Specification

For some requirements in Table 1, Indian Standards do not exist for the test methods, hence reference is provided to other internationally used standards. Once Indian Standards are formulated for these tests, the references will be modified accordingly. Also, alternate test methods are provided below for few characteristics and in case of dispute, the referee methods prescribed in Table 1 shall be followed:

Characteristic	*Alternate Method of Tests*
Density at 15 °C, kg/m^3	ASTM D1657
Vapour pressure at 40 °C, kPa	ASTM D1267, ASTM D6897
Evaporation temperature for 95 percent by volume at 760 mm Hg pressure, °C	EN 15470, EN 15471
Copper strip corrosion at 38 °C for 1 h	ASTM D1838
Hydrogen sulphide	ASTM D2420
Free water content	ASTM D1657 (visual observation)

This standard was first published in 2024. This revision has been brought out to keep pace with the latest technological developments and international practices. In this revision, the following major modifications have been made:

a) The composition, mass fraction of DME has been changed from 20 percent, max to 8 percent, max as an outcome of the R&D Report — 2025 on 'Study of compatibility of elastomers used in LPG distribution chains with various Di Methyl Ether (DME)-LPG blends' issued by LPG Equipment Research Centre, Bangalore, India;

b) Foreword, scope and terminology of Di Methyl Ether (DME) has been modified;

c) Terminology of LPG-DME blend has been incorporated;

(*Continue on third cover*)

Indian Standard

DIMETHYL ETHER (DME) BLENDED LIQUEFIED PETROLEUM GAS (LPG) — SPECIFICATION

(*First Revision*)

1 SCOPE

1.1 This standard prescribes the requirements and methods of sampling and test for Dimethyl Ether (DME) blended Liquefied Petroleum Gas (LPG) marked for distribution and usage in household, commercial and industrial applications excluding automotive use in the country.

1.2 This standard does not purport to address all of the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

2 REFERENCES

The standards listed in Annex A contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are advised to use the latest edition of these standards.

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Liquefied Petroleum Gas (LP Gas or LPG) — *See* IS 4639 (Part 1).

3.1.1 LP gases mainly consist of one or more of the following hydrocarbons:

a) Propane (C_3H_8);

b) Propylene (C_3H_6);

c) *n*-butane (C_4H_{10});

d) ISO-butane (C_4H_{10}); and

e) Butylene (C_4H_8).

3.1.2 Small quantities of one or more of the following hydrocarbons may also be present:

a) Ethane (C_2H_6);

b) Ethylene (C_2H_4);

c) Pentane (C5H12); and

d) Pentene (C_5H_{10}).

3.2 Di Methyl Ether (DME) — Colorless gas containing Oxygen, synthetically produced from gasification of biomass or coal, methanol, Syn gas or natural gas, capable of being used as alternate fuel.

3.3 DME LPG Blend — Blend of LPG and DME as per specified ratio. This blend contains Oxygen molecule.

4 REQUIREMENTS

4.1 LPG shall meet the requirements as per IS 4576.

4.2 DME shall meet the requirements as per IS 16704.

4.3 DME blended LPG shall comply with the requirements given in Table 1 when tested according to appropriate methods given in col (4) of Table 1.

4.4 DME blended LPG shall contain a minimum quantity of odourant, generally a mercaptan, to detect a leak. Adequate odour shall be observed, when tested as mentioned below:

5 ml doctor solution + 8 ml ISO-octane + pinch of sulphur powder to be taken in 25 ml stoppered cylinder. The solution to be shaken and 2 ml LPG (Aq.) to be added. The solution to be shaken slowly by releasing pressure. Odour is adequate if the solution turns yellowish brown.

4.5 Residue of LPG

Subject to agreement between the purchaser and the supplier, the material shall also pass the agreed limits for residue when tested according to IS 1448 (Part 70).

5 PACKING AND MARKING

5.1 Packing

The material shall be packed in suitable cylinders/containers as agreed to between the purchaser and the supplier and subject to the requirements prescribed by statutory authorities. (For cylinders, *see* IS 3196 (Part 1).

5.2 Marking

The cylinders/containers shall be marked as prescribed by statutory authorities from time to time. They shall bear the labels marked with the following information:

a) Name of the material;

b) Mass in kg of the material in the container;

c) Maximum vapor pressure;

d) Manufacturer's name and trade mark, if any; and

e) Any other statutory requirements.

5.2.1 Each cylinder/container shall also be marked with the caution label 'FLAMMABLE' together with the corresponding symbol for labeling dangerous goods [*see* Fig.18 of IS 1260 (Part 1)].

5.2.2 *BIS Certification Marking*

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the standard mark.

6 SAMPLING

6.1 Representative liquid samples of DME blended liquefied petroleum gas shall be drawn as prescribed in IS 1447 (Part 2).

6.2 Tests on vapour pressure shall be conducted on individual samples and the rest of the tests shall be conducted on composite samples.

Table 1 Requirements for DME Blended LPG

(*Clause* 4.3)

Sl No.	Characteristics	Requirements	Method of Test, Ref to
(1)	(2)	(3)	(4)
(i)	Density[1] at 15 ºC, kg/m^3	Report	IS 1448 (Part 76)[8] IS 16074
(ii)	Vapour pressure at 40 °C, kPa, gauge, *Max*	1050 [2]	IS 1448 (Part 71)[8]/ASTM D6897
(iii)	Composition, mass fraction of DME, percent, *m/m*, *Max*	8	IS 18887
(iv)	a) Other Hydrocarbons, liquid volume percent[3]:		
	1) C_2 hydrocarbon;	Report	IS 1448 (Part 151)
	2) C_3 hydrocarbon;	Report	
	3) C_4 hydrocarbon; and	Report	
	4) C_5 hydrocarbon and heavier, *Max.*	2.5	
	OR		
	b) Volatility:	2.2	IS 1448 (Part 72)[8]
	1) Evaporation temperature for 95 percent by volume at 760 mm Hg pressure, °C, *Max*		
(v)	Total volatile sulphur[4], mg/kg (ppmw), *Max*	140	ASTM D6667[8]
(vi)	Copper strip corrosion at 38 °C for 1 h	Not worse than No 1	IS 1448 (Part 152)[8]
(vii)	Hydrogen sulphide[5]	Pass	IS 1448 (Part 73)[8]
(viii)	Free water content[6],[7]	None	IS 1448 (Part 76)[8] (visual observation)
(ix)	Caustic test	Pass	*See* Note 10

NOTES

1 Density/Relative density may also be determined by analyzing the gas by gas chromatograph and using composition and density factor data as per ISO 8973/ASTM D2598/IP 432. DME density at 15 ºC 675 kg/m^3 need to be taken for calculation

2 Each consignment of the product shall be designated by its maximum vapor pressure in kPa at 40 ºC and shall not exceed 16.87 kg/sq.cm at 65 ºC for transportation through road tank trucks. Further, if purchaser and the supplier agreed, the minimum vapor pressure of that mixture shall be not lower than 200 kPa gauge compared to the designated maximum vapor pressures and in any case the minimum for the mixture shall be not lower than 520 kPa at 40 °C.

3 Molar composition of LPG determined by gas chromatographic analysis can be converted into Liquid-Volume using the standard Practice as per ASTM D2421.

4 Total sulphur limits in these specifications do include sulphur compounds used for odorizing purposes.

5 'Pass' the test indicates Hydrogen Sulphide not more than 5 ppm. Hydrogen Sulphide can also be determined with the following procedure:

Take 5 ml of lead acetate solution (prepared in IPA) + 25 ml LPG (liquid) in measuring cylinder and shake well. If black precipitation is not seen, then H_2S test passes.

6 Free water: Water promotes rust on internal surface of steel storage tanks and iron piping systems. Water can also cause odourant to fade and can block small openings. LPG shall not contain any free water at 0 °C and shall be determined by visual inspection of sample using equipment on which density is determined. Subject to agreement between purchaser and supplier, limit, temperature and method for water content determination may be changed for specific applications like shipment of refrigerated LPG, etc.

7 Free water content can also be determined with the following procedure:

a) Take 100 ml of liquid in dry graduated tube and allow to vaporize the LPG. When about 2 ml to 3 ml residue is left, shake and pour the residue on whatman filter paper. Allow the residue on filter paper to evaporate; if wet spot not seen, free water is absent.

8 In case of dispute, this method shall be the referee method.

9 For caustic test following procedure may be adopted:

a) 2 ml LPG (liq) + 5 ml demineralised water in a weathering tube, add a drop of Alizarin Yellow R indicator (Mordant Orange-1, 5-(4-Nitrophenylazo) Salicylic acid, Dye content 70 percent, CAS no 2243-76-7). Caustic test is pass, if it doesn't turns into red from yellow.

b) Preparation of indicator: Take 0.1 gm in 100ml volumetric flask and dissolve it in demineralised water then make up to the mark with demineralised water.

ANNEX A

(*Clause* 2)

LIST OF REFERRED STANDARDS

IS No./Other Standards	*Title*
IS 1260 (Part 1) : 1973	Pictorial marking for handling and labelling of goods: Part 1 Dangerous good (*first revision*)
IS 1447 (Part 2): 2013/ISO 4257 : 2001	Methods of sampling of petroleum and its products: Part 2 Liquefied petroleum gases — Method of sampling (*second revision*)
IS 1448	Methods of test for petroleum and its products
(Part 70) : 2018	Determination of residue in liquefied petroleum gases (*first revision*)
(Part 71) : 2004 /ISO 4256 : 1996	Liquefied petroleum gases — Determination of gauge vapour pressure — LPG method (*second revision*)
(Part 72) : 2023	Volatility test of liquefied petroleum gases (*first revision*)
(Part 73) : 2004 /ISO 8819 : 1993	Liquified petroleum gases — Detection of hydrogen sulphide — Lead acetate method (*first revision*)
(Part 76) : 2019 /ISO 3993 : 1984	Liquified petroleum gases and light hydrocarbons — Determination of density or relative density — Pressure hydrometer method (*first revision*)
(Part 151): 2004 /ISO 7941 : 1998	Commercial propane and butane — Analysis by gas chromatography
(Part 152) : 2004/ISO 6251 : 1996	Liquefied petroleum gases — Corrosiveness to copper — Copper strips test
IS 3196 (Part 1) : 2013	Welded low carbon steel cylinders exceeding 5 litres water capacity for low pressure liquefiable gases: Part 1 Cylinders for liquefied petroleum gases (LPG) — Specification (*sixth revision*)
IS 4576 : 2021	Liquefied petroleum gases — Specification (*fourth revision*)
IS 4639 (Part 1) : 2000/ISO 1998-1 : 1998	Petroleum industry — Terminology: Part 1 Raw materials and product (*first revision*)
IS 16704 : 2018/ ISO 16861 : 2015	Petroleum products — Fuels (Class F) — Specifications of dimethyl ether (DME)
IS 18887 : 2024	DME blended liquified petroleum gas (LPG) fuel — Quantitative determination of dimethyl ether (DME) content by gas chromatography
ISO 8973 : 1997	Liquefied petroleum gases — Calculation method for density and vapour pressure
ASTM D2421-21	Standard practice for interconversion of analysis of C5 and lighter hydrocarbons to gas — Volume, liquid — Volume, or mass basis
ASTM D2598-21	Standard practice for calculation of certain physical properties of liquefied petroleum (LP) gases from compositional analysis
ASTM D6667-21	Standard test method for determination of total volatile sulphur in gaseous hydrocarbons and liquefied petroleum gases by ultraviolet fluorescence
IP 432	Liquefied petroleum gases — Calculation method for density and vapour pressure

ANNEX B

(*Foreword*)

COMMITTEE COMPOSITION

Petroleum and their Related Products of Synthetic or Biological or Natural Origin Sectional Committee, PCD 03

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PCD 3 : 5 /WG 1 - IS 18698 : 2024 - 20 Percent Dimethyl Ether DME Blended Liquefied Petroleum Gas

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Oil Industry Safety Directorate, Noida	SHRI N. K. PANDEY

(*Continue from second cover*)

d) Annex B 'Determination of mass fraction of dimethyl ether (DME) in DME blended LPG' has been substituted by IS 18887; and

e) References have been updated.

The composition of the Committee responsible for the revision of this standard is given in Annex B.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the website-www.bis.gov.in or www.standardsbis.in.

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Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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